



MOTIVATION & OBJECTIVES

Colombia's fintech sector is growing rapidly, yet many institutions lack accessible tools to analyze customer behavior and predict churn. This project aims to:

- Democratize analytics** — enable non-technical stakeholders to explore customer data through interactive visualizations
- Identify behavioral segments** — uncover distinct customer groups via unsupervised clustering on transactional features
- Predict churn risk** — build interpretable models that quantify churn probability and surface key drivers
- Enable what-if analysis** — let decision-makers simulate interventions and assess impact on retention

CORE QUESTION

"How can interactive visual analytics help Colombian fintech institutions understand, segment, and retain their customers?"

DATASET OVERVIEW

The **COFINFAD** (Colombian Fintech Financial Analytics Dataset) contains comprehensive customer and transaction records:

48,723

CUSTOMERS

3.16M

TRANSACTIONS

54

VARIABLES

15

DEPARTMENTS

12 mo

TIME SPAN (2023)

Customer table: demographics, income bracket, acquisition channel, churn probability, NPS, satisfaction score

Transaction log: date, amount (COP), type (Transfer, Withdrawal, Payment, Deposit), behavioral metrics

Feature engineering: 10 behavioral features constructed for clustering — tx_count, avg_tx_value, total_tx_volume, weekend_ratio, tenure, credit_utilization, active_products, satisfaction, spending_volatility, recency_days

METHODOLOGY & APPROACH

We follow a **5-stage analytical framework** that progresses from exploration to actionable simulation:

- EXPLORE** — EDA with interactive charts & maps
- CONFIRM** — Hypothesis testing via ggstatsplot
- SEGMENT** — K-Means, Hierarchical, PAM clustering
- CALIBRATE** — Lasso logistic regression for churn
- SIMULATE** — What-if analysis with real-time sliders

Technology stack:

R Shiny Plotly ggplot2 ggstatsplot Leaflet
glmnet cluster factoextra bslib DT

INTERACTIVE SHINY APPLICATION — 5 ANALYTICAL MODULES

All modules are integrated into a single dark-themed R Shiny dashboard at guofang1028.shinyapps.io/ShinyApp

MODULE 1

Exploratory Data Analysis

Interactive time-series charts, leaflet choropleth maps of regional distribution, and distribution plots by income bracket and transaction type.

[EDA Screenshot]

Replace with Shiny App EDA tab screenshot

MODULE 2

Confirmatory Analysis

Statistical hypothesis testing with ggbetweenstats (t-test, Welch, Mann-Whitney, Kruskal-Wallis) and hclust-reordered correlation matrices.

[CDA Screenshot]

Replace with Shiny App CDA tab screenshot

MODULE 3

Customer Clustering

Multi-algorithm segmentation (K-Means, Hierarchical, PAM) with PCA scatter plots, silhouette validation, and parallel coordinates.

[Clustering Screenshot]

Replace with Shiny App Clustering tab screenshot

MODULE 4

Churn Prediction

Lasso (L1) logistic regression with cross-validation, variable importance ranking, and residual diagnostics.

[Prediction Screenshot]

Replace with Shiny App Prediction tab screenshot

MODULE 5

What-If Simulator

Interactive sliders to adjust customer attributes and observe real-time churn probability changes with sensitivity analysis.

[What-If Screenshot]

Replace with Shiny App What-If tab screenshot

KEY FINDINGS

- Seasonal patterns differ by transaction type** — Transfers peak mid-year while deposits show end-of-year spikes, confirmed by Kruskal-Wallis tests ($p < 0.001$).
- Income brackets drive behavioral divergence** — High-income customers average 3.2x more transaction volume than low-income segments, with significantly different product usage.
- Geographic clustering of churn risk** — Rural departments show 40% higher churn probability than urban centers like Bogotá and Medellín, linked to limited product access.
- Satisfaction score is the #1 churn predictor** — Lasso feature selection consistently ranks satisfaction_score as the strongest predictor, followed by tenure and active_products.
- Product bundling reduces churn ~15%** — What-if simulations show increasing active_products from 1→3 cuts predicted churn from ~47% to ~32%.

DISCOVERED CUSTOMER SEGMENTS

Optimal clustering ($k=4$) reveals four distinct behavioral profiles across 10 engineered features:

Power Users

~18% of customers

High frequency, high value, long tenure. Low churn risk. Cross-sell opportunities.

Regular Users

~35% of customers

Consistent activity, moderate values. Stable segment. Upsell potential via engagement.

Occasional Users

~28% of customers

Sporadic engagement, low product count. Respond well to targeted campaigns.

At-Risk / New Adopters

~19% of customers

Very low activity, short tenure, high churn probability. Priority for onboarding programs.

[Cluster PCA / Heatmap Screenshot]

Replace with clustering visualization screenshot

CONCLUSION & RECOMMENDATIONS

COFINFAD Analytics demonstrates that interactive visual analytics can transform raw fintech data into actionable business intelligence.

- Retain power users** — Deploy loyalty programs with real-time satisfaction monitoring dashboards
- Intervene on at-risk segments** — Personalized outreach and product bundling within 90-day onboarding
- Expand underserved regions** — Region-specific campaigns for rural Colombian departments with high churn
- Leverage what-if simulations** — Use the simulator to quantify ROI of retention initiatives before deployment

FUTURE WORK

- Integrate additional data sources (e.g., RFM scoring, external economic indicators)
- Develop methods to proactively identify network effects among customer clusters
- Expand prediction module with ensemble models (Random Forest, XGBoost)
- Add real-time data streaming for live dashboard monitoring
- Incorporate NLP analysis on support ticket text for deeper churn insights

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